What the calling agent costs

And what to charge for it · as of 13 October 2026 · USD 1 = Rs 278

One minute of a live call costs us Rs 3.25 in Google charges at 40% agent talk time. But the knowledge base is re-sent on every single call, so a one-minute call on the University agent really costs **Rs 8.22**, and a ten-minute one works out at **Rs 3.75 a minute**. Carrier minutes from Infobip are on top and are the one number that could not be proven.

Every figure here is computed from Google's published rates, this account's own Meta billing, and token counts measured on this deployment. Anything uncertain is named in *What could not be proven*.

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1. What one voice call costs us

Google bills a live call in two parts: the audio, charged by the second, and the agent's instructions, charged once when the call connects. The model in use is gemini-3.1-flash-live-preview, at $3.00 per million audio input tokens and $12.00 per million audio output tokens, with audio counted at 25 tokens per second.

Agent talks	Listening	Speaking	Total per minute
30% of the time	Rs 1.25	Rs 1.50	**Rs 2.75**
40% of the time	Rs 1.25	Rs 2.00	**Rs 3.25**
50% of the time	Rs 1.25	Rs 2.50	**Rs 3.75**
60% of the time	Rs 1.25	Rs 3.00	**Rs 4.25**

Listening costs the same whatever happens, because the stream stays open for the whole call. Only the speaking varies.

The instructions cost more than the minutes

The agent's whole prompt — persona, rules and the entire knowledge base — is sent when the call connects. Counted with Google's own token counter against the real assembled prompts:

Agent	Prompt sent per call	Cost	Equals this much talking
University Agent	23,808 tokens	**Rs 4.96**	1.5 minutes
Jazz Agent	3,810 tokens	**Rs 0.79**	0.2 minutes

A complete call, at 40% agent talk time:

Call length	University Agent	per minute	Jazz Agent	per minute
1 minute	Rs 8.22	Rs 8.22	Rs 4.05	Rs 4.05
2 minutes	Rs 11.47	Rs 5.73	Rs 7.30	Rs 3.65
3 minutes	Rs 14.72	Rs 4.91	Rs 10.55	Rs 3.52
5 minutes	Rs 21.23	Rs 4.25	Rs 17.06	Rs 3.41
10 minutes	Rs 37.49	Rs 3.75	Rs 33.32	Rs 3.33

Short calls on a large knowledge base are where the money goes. The Jazz knowledge base is a tenth the size of the University one, and its one-minute call costs barely half as much.

2. What one WhatsApp message costs us

Two separate bills: Meta for delivering the message, Google for writing it.

Meta's charge, from our own billing

These are not quoted rates. They are what Meta actually charged WhatsApp Business Account 2812213305655211, read back from the account over the 90 days to 13 October 2026.

Category	Messages billed	Total charged	Per message
Marketing	2	$0.0946	**Rs 13.15**
Utility	10	$0.1000	**Rs 2.78**
Utility	5	$0.0500	**Rs 2.78**
Service, inside the 24-hour window	7	$0.0000	**Free**

Messages a customer sends us are never charged, and a reply inside 24 hours of their last message is free. That is why auto-reply is cheap and a campaign is not: **one marketing message costs what 4.0 minutes of talking costs.**

Google's charge for writing the reply

The whole knowledge base is sent again on every message, which makes the model choice matter more than anything else in this document.

Text model	University knowledge base	Jazz knowledge base
gemini-3.5-flash (in use now)	**Rs 10.22**	**Rs 1.68**
gemini-3.8-flash	**Rs 5.07**	**Rs 0.80**
gemini-3.5-flash-lite	**Rs 2.08**	**Rs 0.38**

Changing one line of configuration cuts that by 80%. Per 1,000 messages answered on the University knowledge base: Rs 10,219 today against Rs 2,083 on Flash-Lite. The setting is gemini_text_model in backend/app/config.py, and it also drives the Guides assistant and the staff lookup. Flash-Lite is a smaller model, so run scripts/agent_eval.py against the real knowledge bases before switching.

3. The hidden costs

Every one of these is real on this build. Most are invisible until the bill arrives.

What	Why it costs	How much
Silence is billed	The audio stream stays open for the whole call, so input tokens are charged while nobody speaks	Rs 1.25 per minute, whatever happens
The knowledge base is re-sent every time	It is not stored at Google. The whole base goes up on every call and every message	Rs 4.96 per call, Rs 10.22 per message
Thinking tokens are billed as output	The model reasons before answering. Measured on one real reply: 573 thinking tokens against 23 of visible answer	Up to 25x the answer. Now switched off for replies
A call nobody answers still costs	The session connects and the prompt is sent before the phone rings	Rs 4.96 per unanswered call
Marketing templates cost 4.7x utility	Meta's own rate card, confirmed on our account	Rs 13.15 against Rs 2.78
Outside the 24-hour window nothing is free	A free-text reply is accepted and silently not delivered; a paid template is the only way through	Rs 2.78 minimum to reach someone who went quiet
Vercel Hobby forbids commercial use	We are selling this. Hobby is for personal, non-commercial use	Rs 5,560 a month to become compliant
Supabase free pauses after a week idle	A demo company that goes quiet stops working, silently	Rs 6,950 a month to stop it
Render free spins down after 15 minutes	The first inbound call after a quiet spell waits about a minute for the server to wake. The call fails	Lost calls, not rupees

The knowledge base, over a month

It is charged per use, not per month. Trimming it is the cheapest optimisation available — halving the base halves this column.

Calls per month	University base	Jazz base
100	Rs 496	Rs 79
500	Rs 2,482	Rs 397
1,000	Rs 4,964	Rs 794
5,000	Rs 24,820	Rs 3,972

What turned out not to matter

Recordings are cheap. A real call measured 651,260 bytes for 113 seconds, which is 338 KB a minute. Even at 10,000 calls a month averaging four minutes that is 12.9 GB a month, and a year of it still sits inside Supabase Pro's included 100 GB. Playback egress at 20,000 plays a month is 12.9 GB against 250 GB included. **Storage and bandwidth are not worth modelling at our scale.**

4. The monthly bill before anyone makes a call

This is what we pay whether we have one customer or none. We are on free plans today, and two of them we are not entitled to.

Piece	Monthly	Why it has to change
Frontend (Vercel Pro)	Rs 5,560	Hobby is personal, non-commercial use only. We are selling.
Backend (Render Starter)	Rs 1,946	Free spins down after 15 minutes idle and drops inbound calls.
Database and storage (Supabase Pro)	Rs 6,950	Free is 500 MB and pauses after a week of inactivity.
Domain	Rs 278.00	About $12 a year.
Total	**Rs 14,734**	

At about Rs 14,734 a month, roughly $53, the floor is low. It is not where the money goes — usage is.

Render Starter is the urgent one. On the free plan the server sleeps after 15 minutes without traffic and takes about a minute to wake. An inbound call arriving at a sleeping server does not ring; it fails. For a product whose whole promise is answering the phone, that is the most pressing line in the table.

Budget Rs 6,950 a month for Render Standard (2 GB, 1 vCPU) once calls overlap. Each live call holds a WebSocket, an audio pacer and a resampler in memory, and Starter's 512 MB is thin for that. Measure before assuming the step is needed.

5. Where to host it

One constraint rules out most cheap options: **the backend cannot be serverless.** A live call holds an open WebSocket to the carrier and another to Google for its whole length, with an audio clock ticking every 5 ms. Vercel Functions, Lambda and Cloudflare Workers bill per invocation with execution limits and no persistent sockets. The frontend is a static bundle and can go anywhere.

Option	Monthly	For	Against
Render Starter	Rs 1,946	Already deployed, no spin-down, deploys from git	512 MB and 0.5 CPU is thin for concurrent audio
Render Standard	Rs 6,950	2 GB and 1 vCPU, realistic for several calls	3.5x the price of a VPS with the same specs
Hetzner or DigitalOcean VPS	Rs 1,700-3,300	Cheapest per GB of RAM by a wide margin, full control	You run it: updates, TLS, restarts, monitoring, backups
Railway or Fly.io	usage-based	Persistent processes, Fly has regions closer to Pakistan	Usage billing is harder to predict than a flat fee

The recommendation

Stay on Render; move to Starter now and Standard when calls overlap. Not because it is cheapest, because it is not, but because at this stage the difference is about Rs 5,000 a month and a VPS costs a person's attention instead. One unattended reboot during business hours costs more in lost calls than a year of that difference. Revisit when the Render bill passes roughly Rs 20,000 a month.

Keep Supabase. It is the database, the file store and auth in one, and Pro's quotas are far beyond what the measured recordings will use. Replacing it with self-managed Postgres saves about Rs 7,000 a month and costs

backups, pooling and point-in-time recovery.

Move the frontend off Vercel to Cloudflare Pages. Cloudflare allows commercial use on its free tier. That is a real Rs 5,560 a month saved on a plain static build with no server-side logic, so the migration is low risk. It is the one easy saving here.

6. What to charge a company

These are cost floors, not prices. Carrier minutes are missing from every line and must be added once Infobip gives a rate.

Assuming a three-minute average call, 40% agent talk time, and a trimmed knowledge base around the Jazz size:

Package	Calls	Messages	Our cost	Suggested price	Gross margin
Starter	200	500	Rs 7,022	Rs 15,000	53%
Growth	1,000	2,500	Rs 15,464	Rs 45,000	66%
Business	5,000	10,000	Rs 57,672	Rs 165,000	65%

Messages are costed as free service replies inside the 24-hour window, which is what auto-reply actually does. The platform floor is split across three customers; at ten it falls to about Rs 1,473 each.

Three rules the numbers argue for

1. Sell call bundles, not unlimited minutes. Our cost is linear in minutes and the knowledge base makes short calls disproportionately expensive. An unlimited package with a chatty customer loses money.

2. Charge separately for campaigns. Marketing messages at Rs 13.15 each are a pass-through cost, not something a subscription should absorb. A 10,000-contact campaign is Rs 131,494 of Meta charges alone.

3. Cap the knowledge base, or price by its size. A customer who uploads a 200-page manual triples our per-call cost without touching their minute count. Either cap it in the product or make it a pricing tier.

The lever worth pulling first: before raising any price, switch the text model and trim the knowledge bases. Together those cut the Google bill by roughly 70% with no loss of capability we have measured — worth more than any pricing change at this stage.

7. What could not be proven

Four numbers here are not solid. Three are small. One is large enough to change every price above.

Infobip carrier minutes — the missing number

Infobip does not publish per-destination voice rates, and the rate and balance endpoints both return 404 on our api-pk2 instance. Our own call logs came back empty, so there is nothing to read a real price from. Third-party guides put Infobip voice between EUR 0.0002 and EUR 0.0054 a minute without naming Pakistan, which is too wide to build a price on. At the top of that range, about Rs 1.70 a minute, carrier cost would be roughly half what Google charges. Get it from the rate card in the Infobip portal under Billing, or ask the account manager for Pakistan mobile and landline termination, then add it to every per-minute figure here.

The model we run is not on Google's price list

The official pricing page lists Gemini 3.8 Flash and 3.5 Flash-Lite but not gemini-3.5-flash, which is what the build actually uses. The $1.50 and $9.00 rates come from pricing aggregators citing Google's docs, and they agree with each other. It still serves requests. But a model you cannot find on the price list is a model whose price can move without you noticing — a second reason to switch.

Whether Google re-bills the prompt each turn

Our code sends the system instruction once, in the session setup, which is verifiable in gemini.py. Whether Google internally re-charges accumulated context per turn is their implementation and is not documented. If it does, the per-call prompt cost is higher than stated. Reading usage_metadata from one real call settles it, once there is credit on the account.

Render's current prices

Their pricing page would not render, so the $7 and $25 figures come from several independent write-ups that agree with each other. Worth confirming on render.com/pricing before committing.

The exchange rate

Rs 278 to the dollar throughout, the open-market selling rate on 9 October 2026. Every dollar figure moves with it. All of our costs are dollar-denominated while all of our revenue will be in rupees, so a 10% rupee slide takes about seven points off the gross margins above.

8. Sources

Measured on our own systems

Figure	Where it came from
Meta per-message rates	pricing_analytics on WhatsApp Business Account 2812213305655211, 90 days to 13 Oct 2026
Prompt tokens (23,808 and 3,810)	Google's count_tokens run against the assembled prompts on this deployment
Knowledge base sizes (22,706 and 2,238 tokens)	The same, against the stored knowledge bases
Recording size, 338 KB a minute	The one kept production recording: 651,260 bytes for 113 seconds
Thinking tokens, 573 of a 600 budget	usage_metadata from a real reply on the University knowledge base

Figure	Where it came from
Infobip rates unavailable	/voice/1/logs and /account/1/balance both 404; /calls/1/calls returned no results

Published pages

- Gemini Developer API pricing — ai.google.dev/gemini-api/docs/pricing — audio in and out, text in, and the 25 tokens-per-second audio rate
- WhatsApp Business Platform pricing — developers.facebook.com/documentation/business-messaging/whatsapp/pricing — free service window, customer messages never charged
- Supabase pricing — supabase.com/pricing — Free and Pro quotas, overage rates, the one-week pause on free projects
- Vercel pricing — vercel.com/pricing — Hobby free and non-commercial, Pro $20
- Infobip voice pricing — infobip.com/voice/pricing — confirms no published per-destination rates
- Pakistan Observer, 9 October 2026 — USD/PKR open market 277.4 / 278.2

Secondary, treat with care

- gemini-3.5-flash at $1.50 and $9.00 per million — pricepertoken.com and anotherwrapper.com, agreeing, citing Google's docs; not on Google's own page
- Render Starter $7 and Standard $25 — several independent write-ups agreeing; Render's own page would not render

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Every figure in this document is computed by docs/build_cost_report.py rather than typed in. Change the exchange rate, the talk ratio, the call length or a knowledge base size at the top of that file and re-run it, and every table moves with it.